Solid State Physics Practice

Budapest University of Technology and Economics

Problem Set 1

Fall 2026

Keywords: free electron gas, periodic boundary conditions, density of states, Fermi energy, Bravais lattices, primitive vectors, reciprocal lattice, triangular lattice, cuprate CuO_2 plane, crystal symmetry, point groups, conductivity tensor, Neumann's principle

Exercise 1 — Confined Two-Dimensional Electron Gas

Keywords: free electron gas, quantum confinement, density of states, Fermi energy, subbands

Consider non-interacting electrons that are free to move in the x–y plane. For convenience, take a square of side L in the plane and impose periodic boundary conditions along both x and y. In the z-direction, the electrons are confined to an infinite square well of width d,

$$V(z) = \begin{cases} 0, & 0 < z < d, \\ \infty, & \text{otherwise.} \end{cases}$$

Thus, the motion in the z-direction is quantized.

(a) Determine the allowed values of k_x, k_y, and k_z, and find the corresponding single-particle energy eigenvalues. Show that each quantized z-state gives rise to a two-dimensional parabolic subband $E_\nu(k_x, k_y)$, and determine the energy $E_\nu^{(0)}$ at the bottom of each subband.

(b) Derive the density of states $g(E)$ for a single two-dimensional subband, including the spin degeneracy. Compare its energy dependence with that of a three-dimensional free electron gas.

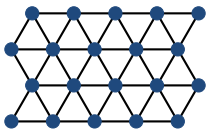
(c) At $T = 0$, first assume that only the lowest subband is occupied. Express the Fermi wavevector k_F and the Fermi energy E_F in terms of the areal electron density n, including the spin degeneracy. Compare the density dependence of E_F with that of a three-dimensional free electron gas.

As the electron density is increased, determine the critical areal density n_c at which the second confinement subband starts to become occupied.

Exercise 2 — Triangular Lattice

Keywords: 2D Bravais lattice, primitive vectors, reciprocal lattice, Brillouin zone

Assume that we have a triangular lattice made of the same atom. Each atom has six equivalent nearest neighbors at distance a, as shown below.



(a) Write down the primitive vectors of the lattice.

(b) Find the reciprocal lattice vectors. What type of lattice is the reciprocal lattice, and how is it oriented relative to the direct lattice?

(c) Sketch (or describe) the shape of the first Brillouin zone.

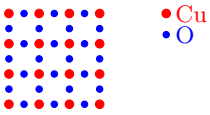
Exercise 3 — Cubic and BCC Lattices

Keywords: Bravais lattice, primitive vectors, reciprocal lattice, simple cubic, body-centered cubic

Give the primitive lattice vectors of (i) the simple cubic and (ii) the body-centered cubic (bcc) lattice. For each case, *explicitly calculate* the reciprocal lattice vectors and identify the resulting reciprocal lattice type.

Exercise 4 — The CuO_2 Plane *Keywords: cuprates, square lattice, unit cell, primitive vectors, reciprocal lattice, 2D symmetry group*

The CuO_2 planes play a central role in the superconductivity of cuprates. Although the superconductivity persists at relatively high temperatures, its origin is still unknown, and it is one of the biggest unsolved problems in physics. The CuO_2 plane is a square lattice of Cu atoms, with O atoms located at the midpoints of each Cu–Cu bond.



(a) Identify the unit cell and write down the primitive lattice vectors of the CuO_2 square lattice.

(b) Calculate the reciprocal lattice vectors. What is the point group of the CuO_2 plane?

Exercise 5 — Conductivity Tensor and Symmetry *Keywords: point group symmetry, Neumann's principle, mirror symmetry, 3-fold rotation, transport tensors*

Consider the conductivity tensor

$$\sigma = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{pmatrix}.$$

Determine its most general form under

(i) mirror symmetry, and

(ii) 3-fold rotational symmetry.